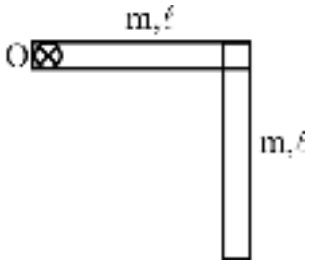
PART 1 : PHYSICS

SECTION-I (i)

1)

Find moment of inertia I of the system (Axis is perpendicular to the the plane passing through O) :



(A) $\frac{16}{3}m\ell^2$

(B) $5m\square^2$

(C) $\frac{2m\ell^2}{3}$

(D) $\frac{5}{3}m\ell^2$

2) Let I_A and I_B be moment of inertia of a body about two axis A and B respectively. The axis of body A passes through the centre of mass of the body but B does not. Then :-

(A) $I_A < I_B$

(B) If $I_A < I_B$, the axes are parallel

(C) If the axes are parallel $I_A < I_B$

(D) If the axes are not parallel $I_A > I_B$

3) In Young's double slit experiment the slits are 0.5 mm apart and interference is observed on a screen placed at a distance of 100 cm from the slits. It is found that the 9th bright fringe is at a distance of 9.0 mm from the second dark fringe from the centre of the fringe pattern. What is the wavelength of light used :-

(A) 2000 Å

(B) 4000 Å

(C) 6000 Å

(D) 8000 Å

4) An electromagnetic wave in vacuum has the electric and magnetic fields $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then :-

(A) $\vec{X} \| \vec{B}$ and $\vec{k} \| \vec{E} \times \vec{B}$

(B) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{B} \times \vec{E}$

(C) $\vec{X} \| \vec{B}$ and $\vec{k} \| \vec{B} \times \vec{E}$

(D) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

SECTION-I (ii)

1) Electrons are used to produce an interference pattern in a standard YDSE setup. Choose the correct statement(s) :-

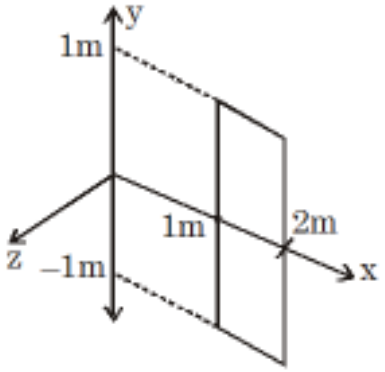
(A) Applying an accelerating voltage between source and slit will result in change in fringe width.

(B) If the size of the slit is increased to few centimeter then fringes will disappear.

(C) No fringe pattern is formed by electrons as they are particles.

(D) Fringe width can be increased by increasing distance between slit & screen or by decreasing distance between the slits.

2) A plane electromagnetic wave $\vec{E} = 3000\frac{N}{c}\hat{j}\sin\left(\pi x - 3\pi \times 10^8 t\right)$ is travelling in vacuum. A conducting loop of dimensions as shown is kept in xy plane as shown. The induced emf in the loop :-



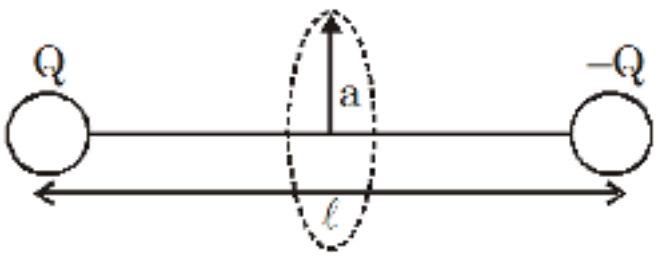
(A) At t = 0 is 0

(B) At t = 0.01 μs is 12000 V

(C) At t = 0.003 μs is 12000 V

(D) At t = $\frac{1}{1800}\mu s$ is 6000 V

3) Consider 2 charged metallic balls of radius r connected by a thin metallic wire. The initial charge on balls is equal and opposite. Assume r << l so that at any instant, the charge on the balls is spherically symmetric. Consider an amperian loop in the symmetry plane of system having radius a as shown.



(A) The displacement current through the loop is towards left and conduction current to right.

(B) The magnetic field at the amperian loop when current in the wire is i is $\frac{\mu_0 i}{2\pi a}$.

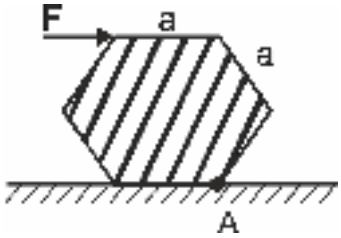
(C) The magnetic field at the amperian loop when current in the wire is i is $\frac{\mu_0 i}{4\pi a}\left(\frac{\ell}{\sqrt{a^2+\frac{\ell^2}{4}}}\right)$.

(D) The magnetic field at the amperian loop when current in the wire is i is zero.

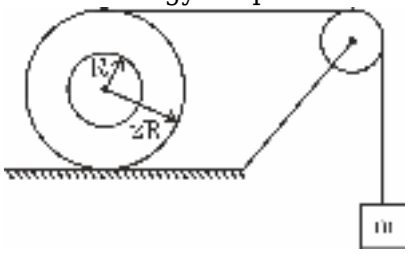
SECTION-II

1) A wheel is rotating about an axis through its centre at 720 rpm. It is acted on by a constant torque opposing its motion for 8 second to bring it to rest finally. Find the value of torque (in N × m) (given moment of inertia of wheel is $I = \frac{24}{\pi} kg - m^2$)

2) A regular hexagonal uniform block of mass m rests on a rough horizontal surface with coefficient of friction μ as shown in figure. A constant horizontal force is applied on the block as shown. If the coefficient of friction is sufficient to prevent slipping before toppling, then the minimum force required to topple the block about its corner A is $\frac{mg}{a\sqrt{b}}$, then 2b/a is ___.



3) In the figure as shown mass of both, the spherical body and block is m. Moment of inertia of the spherical body about COM is $2mR^2$. The spherical body rolls on the horizontal surface. The ratio of kinetic energy of spherical body to that of block at any instant is $\frac{x}{y}$. Then fill the value of x + y.



4) The width of one of the two slits in a Young's double slit experiment is four times the other slit. The ratio of maximum to minimum intensity in the interference pattern is x : 1. Find x.

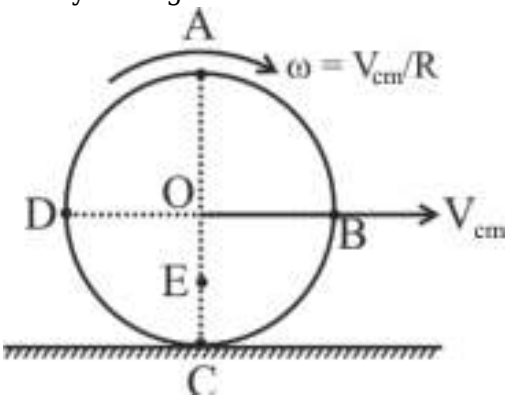
5) A fringe width of 6 mm was produced for two slits separated by 1 mm apart. The screen is placed 10 m away. The wavelength of light used is 'x' nm. The value of 'x' to the nearest integer is _____.

6) An unpolarized light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 units. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of

emerging light will be Nunits. Find N.

SECTION-I (iii)

1) **Answer the following by appropriately matching the lists based on the information given in the paragraph.**
Consider a body rolling on a horizontal surface as shown in figure (Symbols have their usual meaning)



List-I		List-II	
(I)	Velocity is zero	(P)	Point A
(II)	Speed in maximum	(Q)	Point B
(III)	0 < speed < v_{cm}	(R)	Point C
(IV)	1.3 v_{cm} < speed < 2 v_{cm}	(S)	Point D
		(T)	Point E

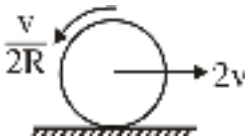
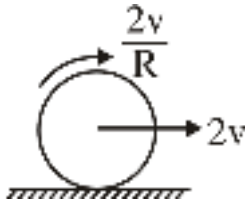
(A) I → P,R;II → Q,S;III → Q;IV → R

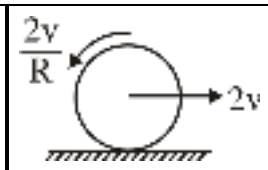
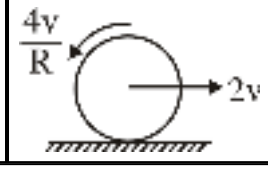
(B) I → Q,S;II → P,T;III → Q,S;IV → Q,R

(C) I → Q,S;II → T;III → P;IV → R

(D) I → R;II → P;III → T;IV → Q,S

2) **Answer the following by appropriately matching the lists based on the information given in the paragraph.** Four different situations of a moving disc are shown in List-I and predictions about its final motion and forces acting on it are given in List-II.

List-I		List-II	
(P)		(1)	Finally disc will roll along the initial direction of velocity (v)
(Q)		(2)	Finally, disc will roll in direction opposite to the initial direction of velocity (v)

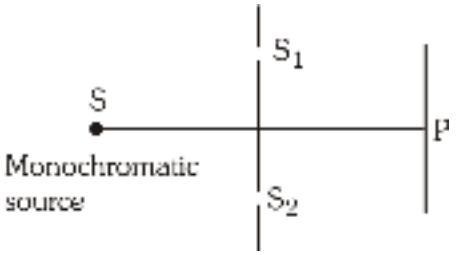
(R)		(3)	Finally, disc stops
(S)		(4)	Initially friction force acts in the direction opposite to that of initial velocity
		(5)	No friction act on the disc.

(A) P → 1,4;Q → 1,5;R → 1,4;S → 3,4

(B) P → 1,3;Q → 2,4;R → 1,4;S → 1,5

(C) P → 1,2;Q → 2,3;R → 3,4;S → 4,5

(D) P → 1,4;Q → 1,4;R → 2,4;S → 2,5



3) A Young's double slit appratus is shown in figure

	List-I		List-II
(P)	Source S move upwards	(1)	Central fringe moves upwards
(Q)	A thin glass slab is placed in front of slit S_2	(2)	Central fringe moves downward
(R)	Whole apparatus is immersed in water	(3)	Fringe width remains constant
(S)	Diameter of slit S_2 is doubled	(4)	Fringe width decreases
		(5)	Intensity of central maxima increases

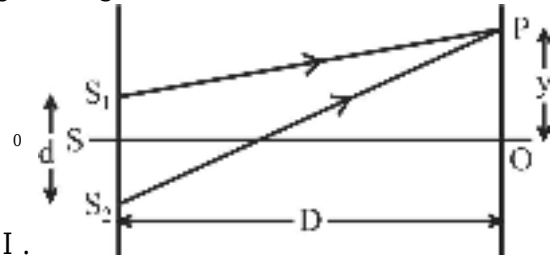
(A) P → 1;Q → 2;R → 3;S → 4

(B) P → 2,3;Q → 1,3;R → 4;S → 5

(C) P → 2;Q → 3;R → 3;S → 4

(D) P → 1;Q → 2;R → 2;S → 3

4) In Young's double-slit experiment; d = 1 mm, wavelength of light is λ = 4000 Å and D = 100 cm.



Now, match the following. intensity of both the source is I .

Column-I		Column-II	
(P)	Fringe width	(1)	0.2 mm

(Q)	If intensity is $2I_0$ at P then y_{min}	(2)	0.000267 mm
(R)	If a glass slab of thickness 0.01 mm and $\mu = 1.5$ is kept on S_1 then shift in fringe pattern	(3)	0.4 mm
(S)	Thickness of slab on S_1 ($\mu = 1.5$) to make intensity at O = I_0	(4)	0.1 mm
		(5)	5 mm

(A) P → 1;Q → 4;R → 2;S → 5

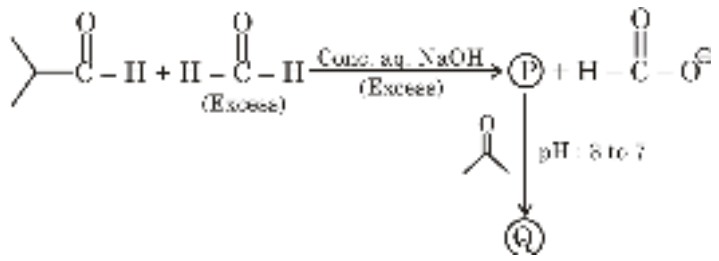
(B) P → 3;Q → 2;R → 1;S → 4

(C) P → 3;Q → 4;R → 5;S → 2

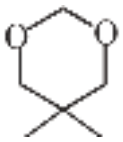
(D) P → 5;Q → 1;R → 3;S → 2

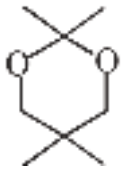
PART 2 : CHEMISTRY

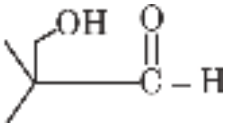
SECTION-I (i)

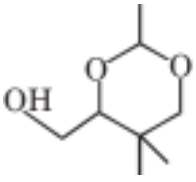


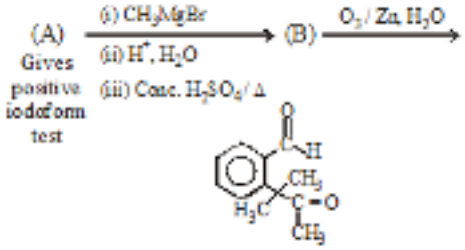
1)
Q is :

(A) 

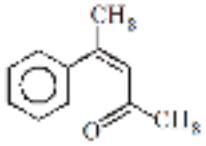
(B) 

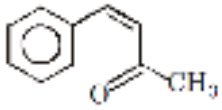
(C) 

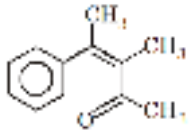
(D) 

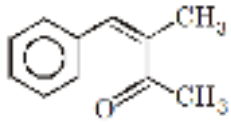


2) Identify (A) in the following reaction sequence :

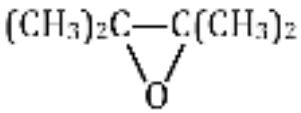
(A) 

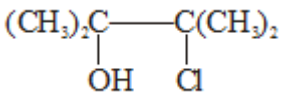
(B) 

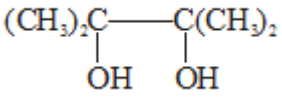
(C) 

(D) 

3) $(CH_3)_2C = C(CH_3)_2 \xrightarrow[H_2O]{X_2} A \xrightarrow[\text{limited amount}]{OH^-} B$, Product 'B' is :

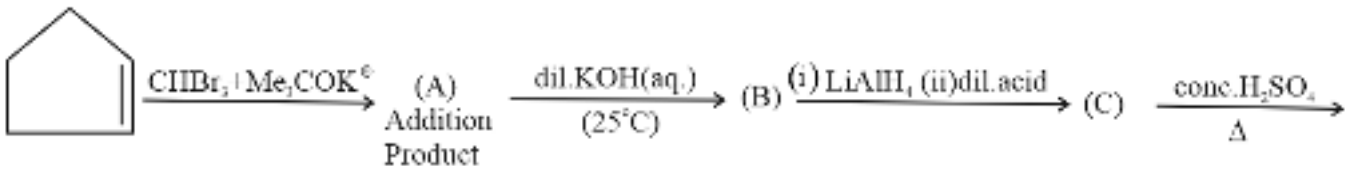
(A) 

(B) 

(C) 

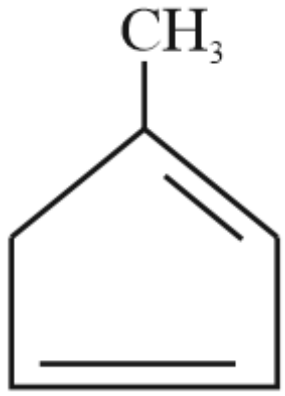
(D) None

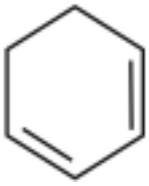
4)

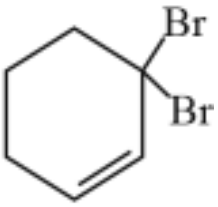


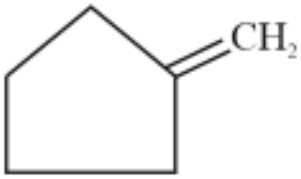
(D)

Identify structure of compound (D).

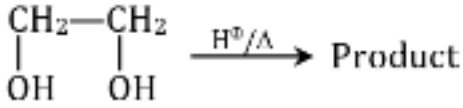
(A) 

(B) 

(C) 

(D) 

SECTION-I (ii)

1) 

Which of the following statement is correct about product?

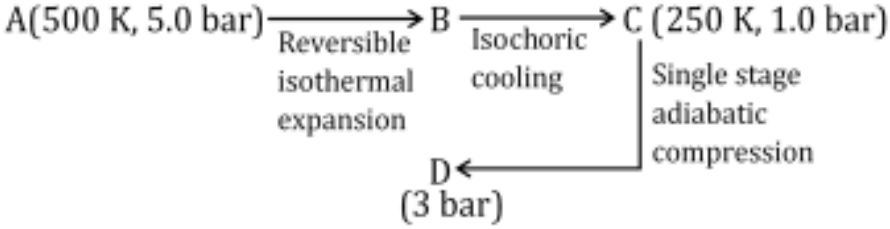
(A) It gives positive tollen's test

(B) It gives positive fehling test

(C) It gives positive lodoform test

(D) It gives positive test with $NaHCO_3$

2) Two moles of an ideal gas ($C_{v,m} = \frac{3}{2}R$) is subjected to following change of state



The correct statement is/are :

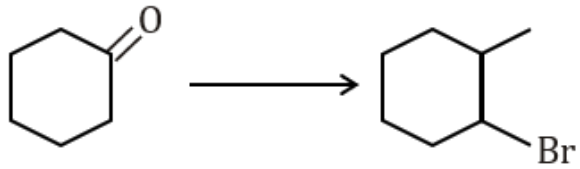
(A) The pressure at B is 2.0 bar

(B) The temperature at D is 450 K

(C) ΔH_{CD} = 1000 R

(D) ΔU_{BC} = 375 R

3) Which combination of reagents will not bring about the following conversion ?



(A) MeMgBr/$H^{\oplus}$, H_2SO_4/Δ, HBr/H_2O_2, hν

(B) MeMgBr/$H^{\oplus}$, H_2SO_4/Δ, HBr

(C) MeMgBr/$H^{\oplus}$, HBr/CCl_4

(D) MeMgBr, NH_4Br

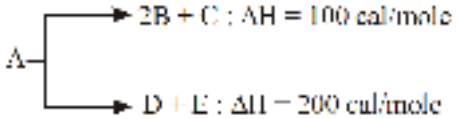
SECTION-II

1) 'X' is a smallest optically active alkanol. On dehydration it can form Y number of alkenes (including stereoisomers). On reaction with Lucas reagent it forms Z number of alkyl halides (including stereoisomers). Report your answer as

Z	Y

2) Find total number of carbonyl compounds with molecular formula $C_7H_{12}O$ which gives Methylcyclohexane on heating with red P + HI?
(NOTE: Each stereoisomer react independently)

3) For the reaction,



If initially 1 mole of A is taken & finally ratio of products B and D is 6 : 1, calculate the enthalpy change (in cal) per mole of A reacted.

4) A non- Allenite who joined Allen test series, appeared in next test. Where he was asked to calculate ΔS_{sys} for a process as described below.
"A dithermic container (containing an ideal gas) fitted with a piston at equilibrium (without any stopper) & has initial volume 600 litre. Now the external pressure is suddenly reduced to 1 bar and allowed the piston to move upward isothermally. In this process system absorbs 600 kJ heat."
He calculated ΔS_{sys} = 2000 J/K but was awarded zero marks. To identify his mistake be contacted his allenite friend and explained his problem. When allenite asked how he calculated ΔS_{sys}, he (non-allenite) said "Simple! by $\frac{q}{T}$". If you are that allenite, then help the non-alleite in getting correct

answer (in J/K) and fill the same in OMR sheet. [ln 11 = 2.4]

5) The minimum pressure (in MPa) require for a given volume of nitrogen gas ($C_{v,m}$ =5/2 R), originally at 300K and 0.10 MPa pressure is x,to be adiabatically compressed in order to raise its temperature to 600K. Then calculate the value of 100x.

6) For silver, $C_p\left(JK^{-1}mol^{-1}\right) = 23 + 0.01\ T$. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH in kJ (nearest integer)

SECTION-I (iii)

1) Match the contents of Column – I with contents of Column – II for 1 mole of gas.

Column - I	Column - I
(A) Raw egg → Boiled egg	(p) ΔH = + Ve
(B) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	(q) ΔS = +Ve
(C) $P_{(white)\ solid} \rightarrow P_{(red)\ solid}$	(r) ΔG = –Ve
(D) Pd + $H_2 \rightleftharpoons$ adsorbed gas	(s) Allotropic change
	(t) ΔS = – Ve

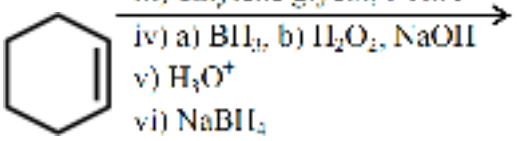
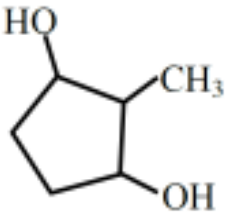
(A) A → r,s,t;B → p,q,r;C → p,r;D → q,s

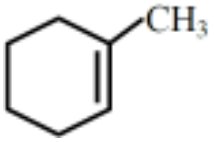
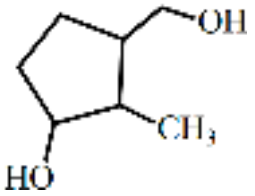
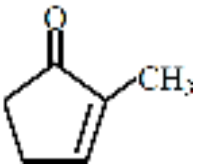
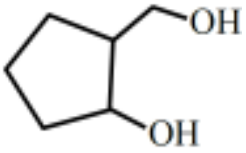
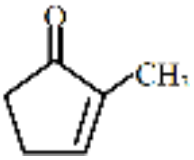
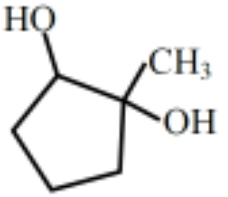
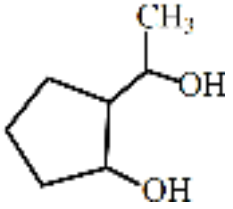
(B) A → p,r,s;B → q,r,t;C → p,q,r;D → p,q,s

(C) A → p,q,r;B → p,q,r;C → p,s;D → p,r,t

(D) A → p,q,s;B → p,r,s;C → q,r,t;D → r,s,t

2) **List-I** contains various reaction sequences and **List-II** contains the possible products. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

	List-I		List-II
(P)	 i) O_3, Zn ii) aq. NaOH, Δ iii) ethylene glycol, PTSA iv) a) BH_3, b) H_2O_2, NaOH v) H_3O^+ vi) $NaBH_4$	(1)	

(Q)	 i) O_3, Zn ii) aq. NaOH, Δ iii) ethylene glycol, PTSA iv) a) BH_3, b) H_2O_2, NaOH v) H_3O^+ vi) $NaBH_4$	(2)	
(R)	 i) ethylene glycol, PTSA ii) a) $Hg(OAc)_2$, H_2O, b) $NaBH_4$ iii) H_3O^+ iv) $NaBH_4$	(3)	
(S)	 i) ethylene glycol, PTSA ii) a) BH_3, b) H_2O_2, NaOH iii) H_3O^+ iv) $NaBH_4$	(4)	
		(5)	

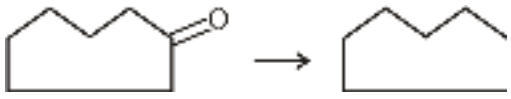
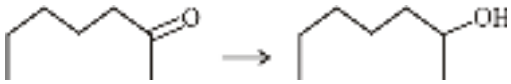
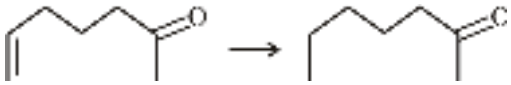
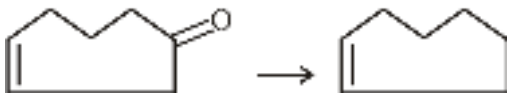
(A) P → 3;Q → 5;R → 4;S → 1

(B) P → 3;Q → 2;R → 4;S → 1

(C) P → 3;Q → 5;R → 1;S → 4

(D) P → 5;Q → 2;R → 4;S → 1

3) Match the list

List-I (Conversion)		**List-II (Reagent)**	
(P)		(1)	Zn / Hg / HCl
(Q)		(2)	$LiAlH_4$
(R)		(3)	NH_2–NH_2 / H_2O_2
(S)		(4)	NH_2–NH_2 / KOH
		(5)	H_2 / Ni

(A) P → 4;Q → 5;R → 5;S → 1

(B) P → 4;Q → 2;R → 3;S → 1

(C) P → 1;Q → 2;R → 3;S → 4

(D) P → 5;Q → 2;R → 4;S → 1

4)

	Column-I (Compound)		Column-II (Properties)
(A)	CH_3–CHO	(P)	Can participate in self Aldol
(B)	HCHO	(Q)	Can participate in haloform
(C)	PhCHO	(R)	Can reduce Tollen's reagent
(D)	CH_3COCH_3	(S)	Can participate in Cannizaro's reaction
		(T)	Can reduce Fehling solution

(A) A → PQRT;B → RST;C → RS;D → PQ

(B) A → PQRT;B → RST;C → PQ;D → ST

(C) A → PT;B → RQ;C → QS;D → PQ

(D) A → RST;B → PQR;C → QST;D → ST

PART 3 : MATHEMATICS

SECTION-I (i)

1) If $\frac{a}{\sqrt{bc}} - 2 = \sqrt{\frac{b}{c}} + \sqrt{\frac{c}{b}}$, where a, b, c > 0, then the family of lines $\sqrt{a}x + \sqrt{b}y + \sqrt{c} = 0$ passes through the fixed point given by:

(A) (1, 1)

(B) (1, –2)

(C) (–1, 2)

(D) (–1, 1)

2) Let P be (5, 3) and a point R on y = x and Q on the x–axis be such that PQ + QR + RP is minimum. Then the coordinates of Q are

(A) $\left(\frac{17}{4}, 0\right)$

(B) (17, 0)

(C) $\left(\frac{17}{2}, 0\right)$

(D) None of these

3) Solution of differential equation $x^2 \frac{dy}{dx} + y^2 e^{\frac{x(y-x)}{y}} = 2y(x-y)$ be given by
Where C is constant of integration

(A) $x(x + y) = y \log (Ce^x - 1)$

(B) $x(x - y) = x \log (Ce^x - 1)$

(C) $x(x + y) = x\log (Ce^x + 1)$

(D) $x(x - y) = y \log (Ce^x - 1)$

4) If $f(x) = \sin x \ \forall\, x \in \left[0, \frac{\pi}{2}\right], f(x) + f(\pi - x) = 2 \ \forall\, x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x) \forall\, x \in (\pi, 2\pi]$ then the area enclosed by y = f(x) and x-axis is :

(A) π

(B) 2π

(C) 2

(D) 4

SECTION-I (ii)

1) Consider three distinct lines $x + \lambda y + 6 = 0$, $2x + y - 3 = 0$ & $\lambda x + 2y + 5 = 0$ let m denotes number of possible real values of λ for which given lines are concurrent and n denotes number of possible real values of λ for which given lines do not form a triangle, then

(A) m = 2

(B) m = 3

(C) n = 6

(D) n = 7

2) For any real numbers α and β, let $y_{\alpha, \beta}(x)$, $x \in R$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = xe^{\beta x}$, $y(1) = 1$.
Let $S = \{y_{\alpha, \beta}(x) : \alpha, \beta \in R\}$. Then which of the following functions belong(s) to the set S?

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

3) Let 'a' be a positive constant number. Consider two curves $C_1 : y = e^x$, $C_2 : y = e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis, then -

(A) $\lim_{a \to -\infty} S = 1$

(B) $\lim_{a \to 0} \frac{S}{a^2} = \frac{1}{4}$

(C) Range of S is $[0, \infty)$

(D) S(a) is neither odd nor even

SECTION-II

1) Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line x = 1 and D_2 denotes the region surrounded by C_1, C_2 and the line x = a. If $D_1 = D_2$, then the value of $\frac{a}{e}$ is

2) A farmer F_1 has a land in the shape of a triangle with vertices at P(0, 0), Q(1, 1) and R(2, 0). From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n (n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is

3) If y(x) is the solution of the differential equation $\frac{dy}{dx} = -2x(y-1)$ with y(0) = 1, then $\lim_{x\to\infty} y(x)$ equals

4) If the curve satisfying $xdx = \left(\frac{x^2}{y} - y^3\right) dy$ passes through (0, 2) then value of $(y(4))^2\left(4-(y(4))^2\right)$ is M^2, then |M| is

5)

Let P be any point of the line x – y + 3 = 0 and A be a fixed point (3, 4). If the family of lines given by the equation (3 sec θ + 5 cosec θ) x + (7 sec θ – 3 cosec θ) y + 11 (sec θ – cosec θ) = 0 are concurrent at a point B for all permissible values of θ and maximum value of |PA – PB| = $2\sqrt{2n} (n \in N)$ then find the value of n.

6) The number of integral values of parameter α, so that the point $(\alpha, (1+\alpha^2)^{-1})$ does not lie outside the triangle formed by the lines $L_1 : 15y = x + 1$, $L_2 : 78y = 118 - 23x$ and $L_3 : y + 2 = 0$, is equal to

SECTION-I (iii)

1)

List-I		List-II	
(P)	The joint equation of lines x + y = 1 and x – y = 4 is $x^2 - y^2 = ax + by + c$ then a + b + c is	(1)	8
(Q)	If slope of one of the lines represented by equation $ax^2 - 6xy + y^2 = 0$ is square of other then a is	(2)	1

(R)	The joint equation of the lines passing through (0, 0) and perpendicular to the lines given by $2x^2 - 5xy + y^2 = 0$ is $ax^2 + 2hxy + by^2 = 0$, then a + 2h + b is	(3)	4
(S)	If angle between 2 lines represented by $2x^2 + 5xy + 3y^2 + 7y + 4 = 0$ is $\tan^{-1}(m)$ then 5 m is	(4)	–24
		(5)	0

(A) P → 3;Q → 2;R → 1;S → 4

(B) P → 3;Q → 1;R → 1;S → 2

(C) P → 2;Q → 1;R → 4;S → 5

(D) P → 2;Q → 1;R → 1;S → 4

2) Match List-I with List-II and select the correct answer using the code given below the list.

List-I		**List-II**	
(P)	If function $y = e^{4x} + 2e^{-x}$, is a solution of the differential equation $\frac{\frac{d^3y}{dx^3} - 13\frac{dy}{dx}}{y} = K$, then value of $\frac{K}{3}$ is	(1)	3
(Q)	Number of straight lines which satisfy the differential equation $\frac{dy}{dx} + x\left(\frac{dy}{dx}\right)^2 - y = 0$ is	(2)	4
(R)	If real value of m for which the substitution $y = u^m$ will transform the differential equation $2x^4y\frac{dy}{dx} + y^4 = 4x^6$, into a homogeneous equation then '2m' is	(3)	2
(S)	If solution of differential equation $x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 12y$ is, $y = Ax^m + Bx^{-n}$, (m > 0, n < 0) then |m + n| is	(4)	0

(A) P → 4;Q → 1;R → 2;S → 3

(B) P → 3;Q → 2;R → 4;S → 1

(C) P → 2;Q → 3;R → 1;S → 4

(D) P → 1;Q → 2;R → 4;S → 3

3) Column-I contains some differential equations & Column-II contains their general or particular solution. The letter 'C' appearing in Column-II is arbitrary constant.

List-I		**List-II**	
(P)	A solution of differential equation $y\left(2x^4 + y\right)\frac{dy}{dx} = \left(1 - 4xy^2\right)x^2$ is	(1)	$y^2 = (\square nx)^2 + C$

(Q)	A solution of differential equation $\frac{\sum_{n=1}^{\infty}\frac{x^{2n-1}}{(2n-1)!}}{\sum_{n=1}^{\infty}\frac{x^{2n-2}}{(2n-2)!}}=\frac{dx-dy}{dx+dy}$, $n \in N$ is	(2)	$3(x^2y)^2 + y^3 - x^3 = C$
(R)	A solution of differential equation $x=\sum_{r=0}^{\infty}\frac{(xy)^r}{r!}\left(\frac{dy}{dx}\right)^r$ is	(3)	$e^{x^2}\left(y^2-1\right)+e^{y^2}=C$
(S)	A solution of differential equation $\left(e^{x^2}+e^{y^2}\right)y\frac{dy}{dx}+e^{x^2}\left(xy^2-x\right)=0$ is	(4)	$2ye^{2x} = Ce^{2x} - 1$
		(5)	$y = 1$

(A) P → 2;Q → 4;R → 1,2;S → 3

(B) P → 1,2;Q → 3,4;R → 1;S → 5

(C) P → 2,3;Q → 4;R → 1,2;S → 3

(D) P → 2;Q → 4;R → 1;S → 3,5

4) Let $S_1 \equiv [f(x)]^{1/3} - [f(y)]^{2/3} = 0$ and $S_2 \equiv [f(x)]^{2/3} + [f(y)]^{2/3} = 12$ be two curves, satisfying the relation $[f(x + y) - f(x - y)]x - [f(x - y) + f(x + y)]y = 4xy(x^2 - y^2)$

	List-I		**List-II**
(P)	$\int_0^1 f(x)\,dx$ is equal to	(1)	$2\pi + \sqrt{3}$
(Q)	Area bounded by S_1 and S_2 is	(2)	$\frac{1}{4}$
(R)	Area bounded by S_2 $\lvert x\rvert + \lvert y\rvert = \sqrt{12}$ is	(3)	$12\pi - 24$
(S)	Area bounded by S_1 and $y = x$ is	(4)	$\frac{1}{6}$
		(5)	$10\pi - \sqrt{2}$

(A) P → 4;Q → 3;R → 1;S → 4

(B) P → 4;Q → 1;R → 3;S → 2

(C) P → 2;Q → 1;R → 3;S → 4

(D) P → 2;Q → 3;R → 1;S → 4

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	**D**	**C**	**C**	**D**

SECTION-I (ii)

Q.	5	6	7
A.	**A,B,D**	**A,D**	**A,C**

SECTION-II

Q.	8	9	10	11	12	13
A.	**72**	**3**	**11**	**9**	**600**	**75**

SECTION-I (iii)

Q.	14	15	16	17
A.	**D**	**A**	**B**	**C**

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	18	19	20	21
A.	**B**	**D**	**A**	**B**

SECTION-I (ii)

Q.	22	23	24
A.	**A,B,C**	**A,B,C**	**B,C,D**

SECTION-II

Q.	25	26	27	28	29	30
A.	**23**	**6**	**125**	**5280**	**45**	**62**

SECTION-I (iii)

Q.	31	32	33	34
A.	**C**	**A**	**C**	**A**

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	35	36	37	38
A.	**D**	**A**	**D**	**B**

SECTION-I (ii)

Q.	39	40	41
A.	**A,C**	**A,C**	**A,B,C,D**

SECTION-II

Q.	42	43	44	45	46	47
A.	**1**	**4**	**1**	**4**	**5**	**4**

SECTION-I (iii)

Q.	48	49	50	51
A.	**B**	**C**	**D**	**C**

SOLUTIONS

PART 1 : PHYSICS

1) **Question Explanation:**
We need to find the moment of Inertia I of the system about an axis perpendicular to the plane and passing through point O.

Concept:
This question is based on moment of Inertia, parallel axis theorem.

Solution:
$$\left(\frac{m\ell^2}{3}\right) + \left(\frac{m\ell^2}{12} + mx^2\right)$$
where $x^2 = \square^2 + \square^2/4$

Final Answer:
The correct option is (4)

2) I_{CM} is the minimum possible M.I. of body,
$I_d = I_{CM} + Md^2$
For I_{min}, d = 0
When axis are parallel.

3) **Question Explanation :** To find the wavelength of light when gap between one of the bright and one of the dark fringe is given.
Concept :
This question is based on YDSE

Solution :
The distance of the mth bright fringe from the central fringe is
$$y_m = \frac{m\lambda D}{d} = m\beta$$
where $\beta = \lambda D/d$ is the fringe width
$\square$ $y_9 = 9\beta$ (i)
The distance of the mth dark fringe from the central fringe is
$$y'_m = \left(m - \frac{1}{2}\right)\frac{\lambda D}{d} = \left(m - \frac{1}{2}\right)\beta$$
$\square$ $y'_2 = \frac{3}{2}\beta$ (ii)
From Eqs. (i) and (ii), we get $y_9 - y'_2$
$$= 9\beta - \frac{3}{2}\beta = \frac{15}{2}\beta$$
It is given that $y_9 - y'_2 = 9.0$ mm and
$$\beta = \frac{9.0 \times 2}{15} = 1.2\text{mm}$$

$= 1.2 \times 10^{-3}$ m

Now $\lambda = \beta d/D$. Substituting for β, d and D, we get

$\lambda = 6 \times 10^{-7}$ m = 6000 Å

Hence,
Option (3) is correct.

4) $\vec{X} \| \vec{E}$

The direction of wave propagation is parallel to

$\vec{k} \| \vec{E} \times \vec{B}$

5) Electron will behave just like a wave (matter wave). Applying electric field between the screen & slit will result in change in wave-length. If opening is of the order of few centimeter, particle nature of electron dominate & thus no interference.

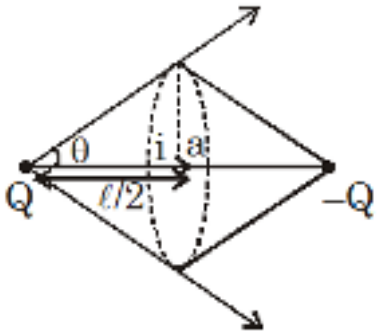
6) $\vec{B} = \frac{E}{C}\hat{k}\sin\left(\pi x - 3\pi \times 10^8 t\right) = 10^{-5}T\hat{k}\left(\sin(\pi x - \omega t)\right)$

$$\phi = \int BdA = 2\int_1^2 Bdx$$

$$= 2 \times 10^{-5}\left[\int_1^2 \sin(\pi x - \omega t)\,dx\right]$$

$$= 2 \times 10^{-5}\left[-\cos\frac{(\pi x - \omega t)|_1^2}{\pi}\right] = \frac{-2 \times 10^{-5}}{\pi}[2\cos\omega t] = -\frac{4}{\pi} \times 10^{-5}\cos\omega t$$

$$r = \frac{d\phi}{dt} = -\frac{4}{\pi} \times 10^{-5} \times 3\pi \times 10^8 \sin\omega t = -12000 \sin \omega t$$



7)

$$\phi_E = \frac{Q}{2\epsilon_0}(1 - \cos\theta) \times 2$$

$$\epsilon_0\frac{d\phi_E}{dt} = i_d \; ; \; i_d = \frac{dQ}{dt}(1 - \cos\theta)$$

Since Q is decreasing = i(1 – cos θ)

i_d is to left.

$$\oint \vec{B}.d\vec{\ell} = \mu^0(i + i_d) = \mu^0[i - i(1 - \cos\theta)]$$

$B \times 2pa = \mu_0 i\cos\theta$

$= \mu_0 i \times \frac{\ell/2}{\sqrt{a^2 + \frac{\ell^2}{4}}}$; $B = \frac{\mu_0 i \ell}{4\pi a\sqrt{a^2 + \frac{\ell^2}{4}}}$

8) $\omega_o = \frac{2\pi N}{60} = \frac{2\pi \times 720}{60} = 2\pi \times 12 = 24\pi$ rad/sec

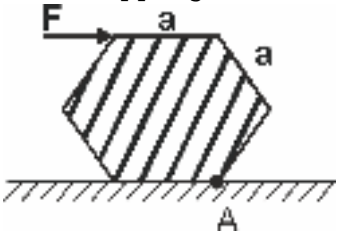
$\omega = \omega_0 + \alpha t$

$0 = (24\pi - \alpha \times 8)$

$\alpha = 3\pi$

$\tau = I\alpha = \frac{24}{\pi} \times 3\pi = 72$ Nm

9) For toppling of block about A



$\Rightarrow \tau_{net} = 0$

$\Rightarrow F.\sqrt{3}a = mg.\frac{a}{2} \Rightarrow F = \frac{mg}{2\sqrt{3}}$

10) $\frac{\frac{1}{2}mv^2 + \frac{1}{2}2mR^2\frac{v^2}{4R^2}}{\frac{1}{2}m(2v)^2} = \frac{3}{8}$

11) $\frac{\left(\sqrt{4I_0} + \sqrt{I_0}\right)^2}{\left(\sqrt{4I_0} - \sqrt{I_0}\right)^2} = \frac{9}{1}$

12) $\beta = \frac{\lambda D}{d}$

$\lambda = \frac{\beta d}{D}$

$\lambda = \frac{6 \times 10^{-3} \times 10^{-3}}{10}$

$\lambda = 6 \times 10^{-7}$ m $= 600 \times 10^{-9}$ m

$\lambda = 600$ nm

13)

After passing from polarizer and analyzer measured intensity = $\frac{I_0}{2} = 100$ Lumens.

After rotating analyzer

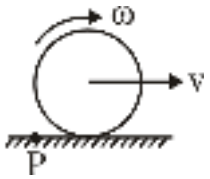
emerging intensity $(I) = \frac{I_0}{2}\cos^2\theta$

$$I = 100\cos^2 30^\circ = 100\left(\frac{\sqrt{3}}{2}\right)^2 = 75$$ Lumens

14) C is ICOR & speed of point at a distance r from C is $\omega r = \frac{v_{CM} r}{R}$

15)

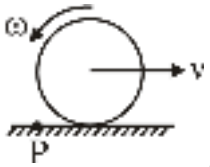
Case-I



about point P angular momentum will be conserved as net torque on the disc is zero.

$L_P = mvR + \frac{mR^2}{2}\omega$ (clock wise)

Case-II



$L_P = mvR - \frac{mR^2}{2}\omega$ (clockwise)

If angular momentum comes out clockwise then it moves in the same direction of initial velocity if it is zero then finally it stops and if it is anticlockwise then it moves in left direction.

16) Conceptual

17)

$W = \frac{D\lambda}{d}$

$= \frac{1 \times 4000 \times 10^{-10}}{1 \times 10^{-3}}$

$= 4 \times 10^{-4}$ meter

= 0.4 mm

$\Delta x = d \sin\theta$

$= d \tan\theta$

$= \frac{dy}{D}$

$I = 2I^0 = 4I^0 \cos^2\left(\frac{\phi}{2}\right)$

$\cos^2\left(\frac{\phi}{2}\right) = \frac{1}{2}$

$\cos\left(\frac{\phi}{2}\right) = \frac{1}{\sqrt{2}}$

$\phi = \frac{\pi}{2}$

$= \frac{2\pi}{\lambda} \times \frac{dy}{D}$

$y = \frac{D\lambda}{4d}$

$= \frac{0.4}{4}$

= 0.1 mm

Shift in fringe pattern,

$x = \frac{D}{d}(\mu - 1)t$

$= \frac{1 \times (1.5 - 1)}{1 \times 10^{-3}} \times 0.01 \times 10^{-3}$

= 0.005 meter

= 5 mm

For intensity,

$I^0 = 4I^0 \cos^2\left(\frac{\phi}{2}\right)$

$\cos^2\left(\frac{\phi}{2}\right) = \frac{1}{2}$

$\phi = \frac{2\pi}{3}$

$\Delta x = \frac{\lambda}{2\pi} 4\phi$

$= (\mu - 1)t$

$= \frac{\lambda}{2\pi} \times \frac{2\pi}{3}$

$= (1.5 - 1)t$

$\Rightarrow t = \frac{\lambda}{3} \times 2 = \frac{2\lambda}{3}$

$t = \frac{2 \times 4000}{3} \times 10^{-10}$

$= \frac{8}{3} \times 10^{-7}$ meter

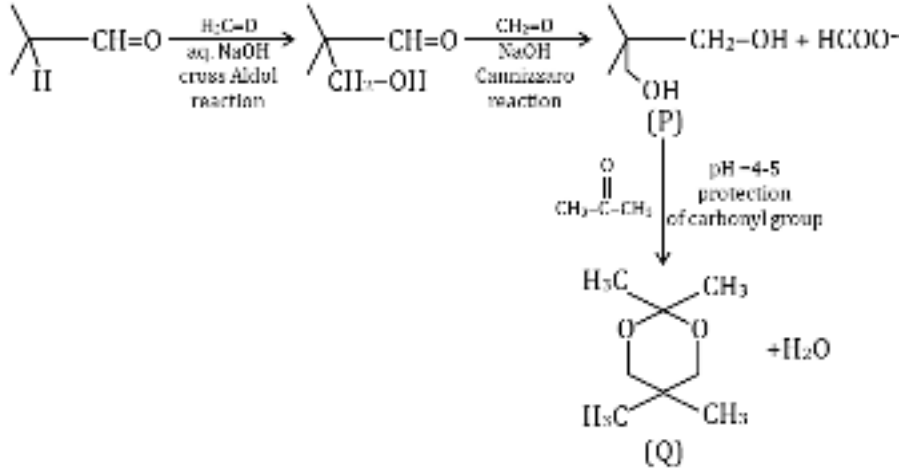
$= \frac{8}{3} \times 10^{-4}$ mm

$= 2.67 \times 10^{-4}$ mm

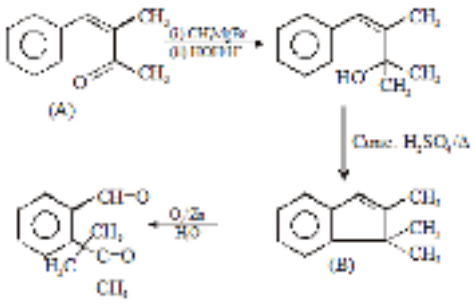
= 0.000267 mm

PART 2 : CHEMISTRY

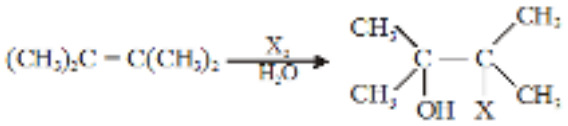
18) **Solution/Explanation/Calculation :**



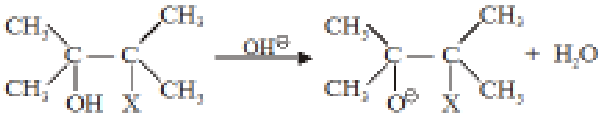
19)



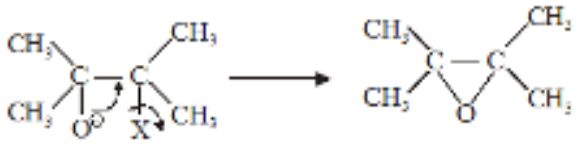
20)



(i) The above step is electrophilic addition reaction on alkene

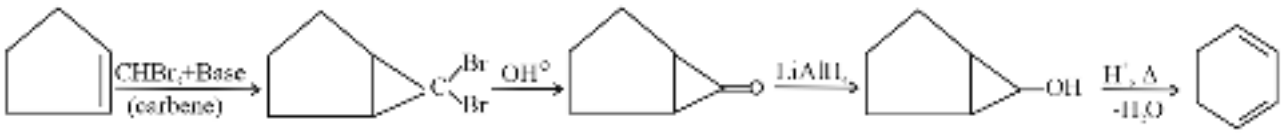


(ii) acid-base reaction

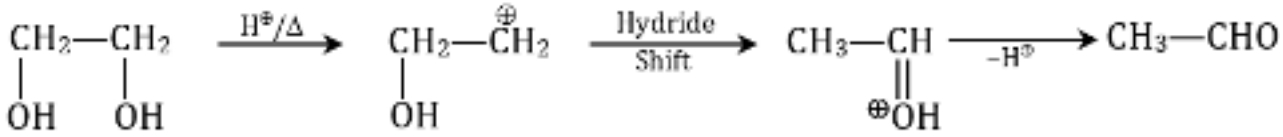


(iii) intromolecular reaction

21)



22)



23) $\frac{P_B}{T_B} = \frac{P_C}{T_C}$

$P_B = \frac{1 \times 500}{250} = 2$ bar

$\Delta U = W$

$nC_v(T_2 - T_1) = -P[V_2 - V_1]$

For process CD

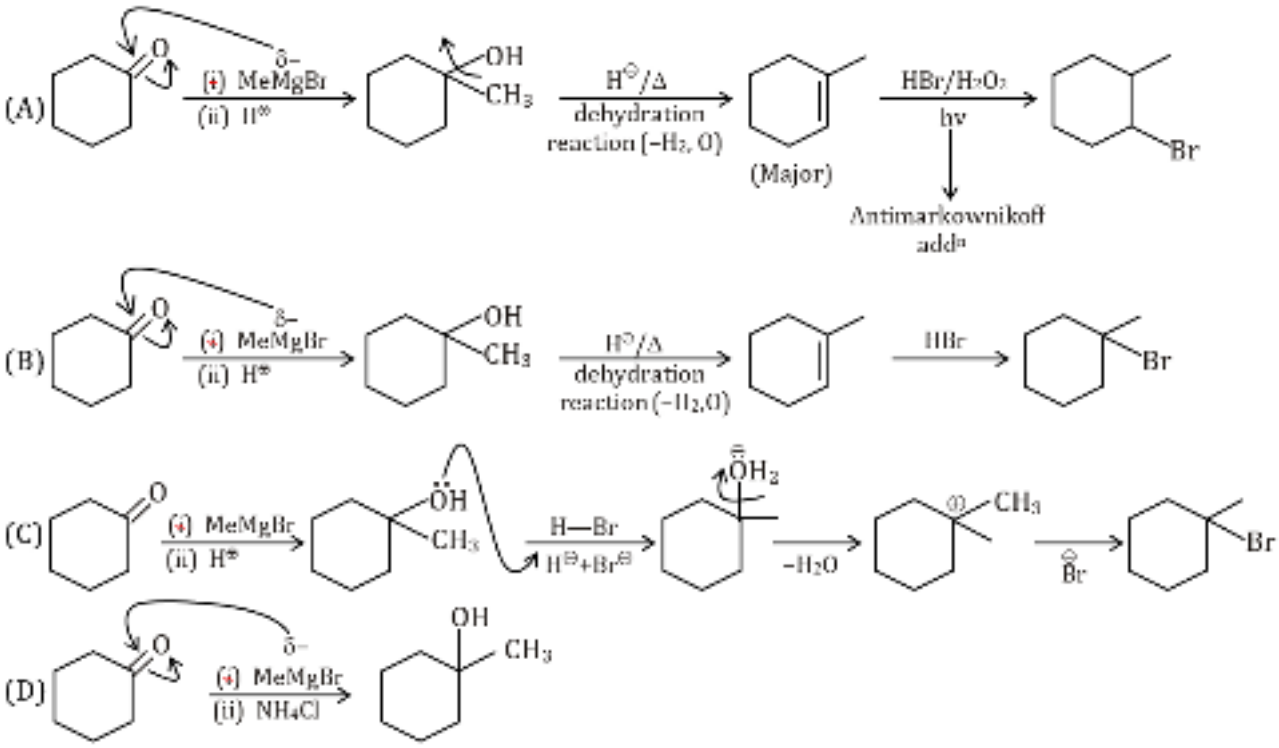
$nC_V(T_D - T_C) = -Pext\left[\frac{nRT_D}{P_D} - \frac{nRT_C}{P_C}\right]$

$2 \times \frac{3}{2}R(T_D - T_C) = -3\left(\frac{T_D}{3} - \frac{T_C}{1}\right) \times 2 \times R$

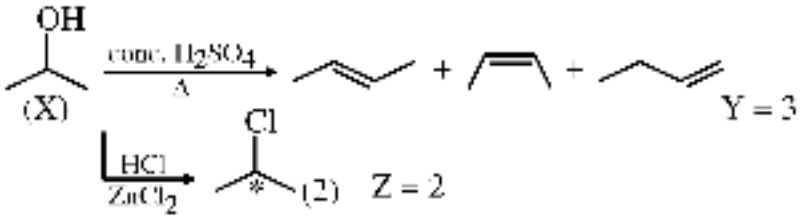
$T_D = 450K$

Also, $\Delta H_{CD} = nC_p\Delta T = 2 \times \frac{5}{2}R \times 200 = 1000R$

24)



25) **New Solution :**



2	3

Old Solution :

Question

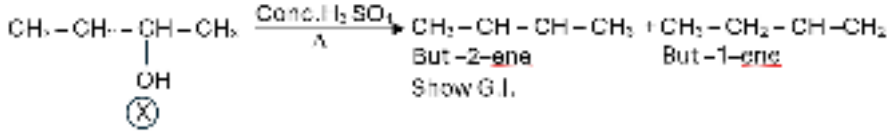
Integer type question :- ZY

Concept

Dehydration of alcohol and reaction of alcohol with lucas reagent.

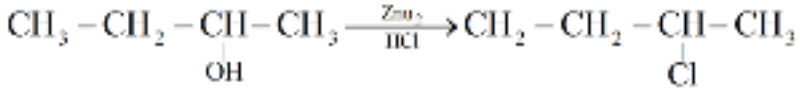
Solution

smallest optically active alcohol is :- Butan–2–01



[Butan–2–01] [Hence cis & Trans] forms are formed

⇒ ∴ Number of Alkenes including stereo isomers ⇒ 3



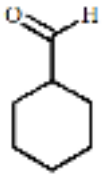
2-chloro butane

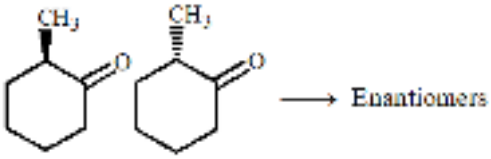
[it is one chiral carbon hence two product are formed]

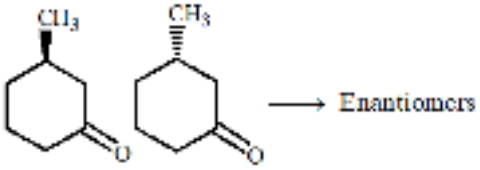
⇒ ∴ Number of alkyl halide formed are 2 including stereo isomers.

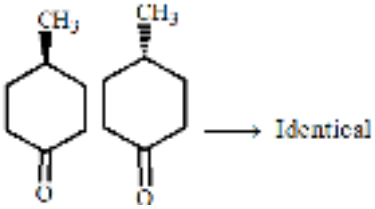
Conclusion

Hence, option Ans 23 is correct







26) 

27)

$A \rightarrow 2B + C$, $\Delta H = 100$ cal / mol

x 2x

$A \rightarrow D + E$, $\Delta H = 200$ cal / mol

1– x 1 – x

$$\frac{n_B}{n_D} = \frac{2x}{1-x} = \frac{6}{1}$$

$x = 3 - 3x$

$$x = \frac{3}{4}$$

in 1st & 2nd reaction enthalpy change of A

$$\Delta H = \frac{3}{4} \times 100 + \frac{1}{4} \times 200$$

= 75+ 50 = 125 cal/mol.

28) $\Delta S_{sys(wrong)} = 2000 = \dfrac{600 \times 1000}{T}$

T = 300 K

$2 = -w = + P(V_2 - V_1)$

$600 \times 1000 = + 1 (V_2 - 600) \times 100$

$V_2 = 6600$ l

$$nR = \frac{P_2V_2}{T} = \frac{1 \times 6600}{300} \times 100 \text{ J}$$

= 2200 J

$(\Delta S)Sys_{(correct)} = nR \ln \frac{V_2}{V_1}$

$= 2200 \ln \frac{6600}{600}$

$= 2200 \times 2.4 = 5280$ J

29) For minimum pressure, the process should be irreversible. Now,

$\Delta U = w$

or, $n \cdot C_{v,m} \cdot (T_2 - T_1) = - P_{ext}(V_2 - V_1)$

or, $n \frac{5}{2} R. (600 - 300)$

$= - P_2 \left(\frac{nR.600}{P_2} - \frac{nR.300}{P_1} \right)$

$\Rightarrow P_2 = 0.45$ MPa $= x$

$\Rightarrow 100x = 45$

30)

Given, $n = 3$ moles, $T_1 = 300$ K, $T_2 = 1000$ K, $C_p = 23 + 0.01T$

On substituting the given values we get

$$\Delta H = 3\int_{300}^{1000}(23 + 0.01T)dT = 3\int_{300}^{1000}23dT + 0.01T\,dT$$

$$= 3\left[23T + \frac{0.01T^2}{2}\right]_{300}^{1000}$$

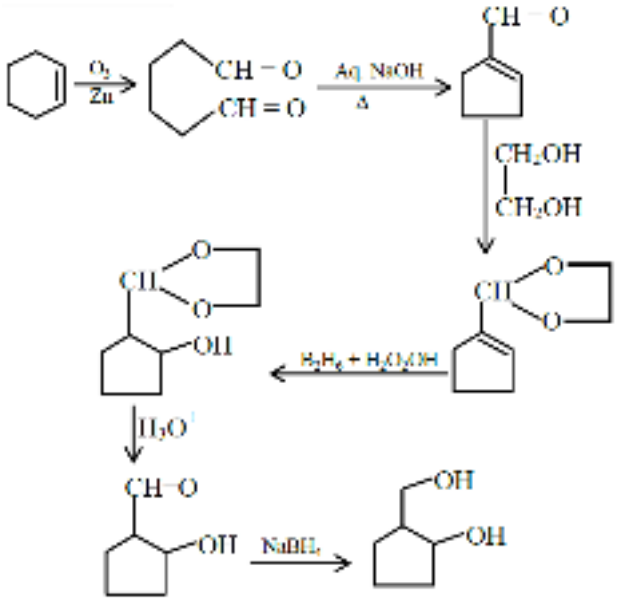
$$= 3\left[23(1000 - 300) + \frac{0.01}{2}(1000^2 - 300^2)\right]$$

$$= 3[16100 + 4550] = 61950 \text{ J} \approx 62 \text{ kJ}$$

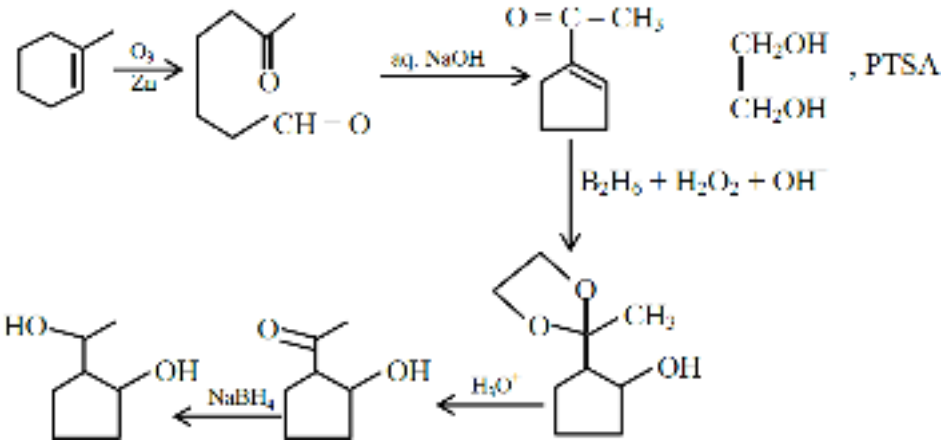
31)

(A) → p,q,r
(B) → p,q,r
(C) → p,s
(D) → p,r,t

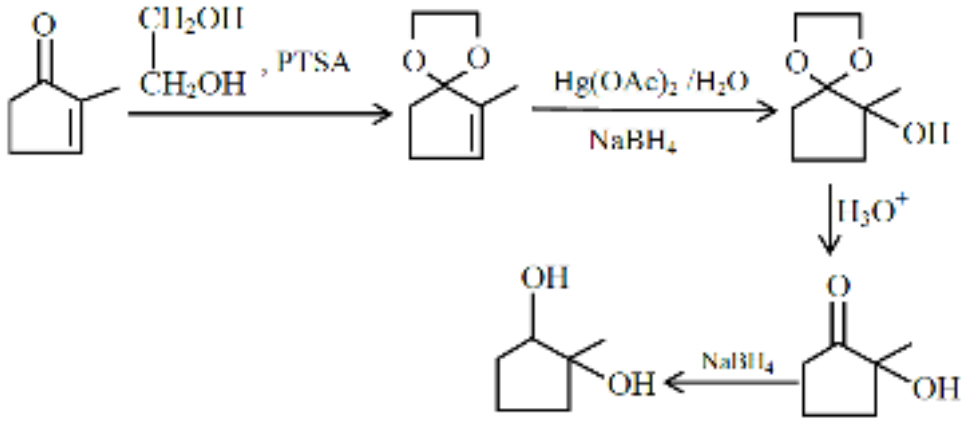
32) (P)


O3
Zn
CH – O
CH = O
Aq. NaOH
Δ
CH – O
CH2OH
CH2OH
CH
O
O
OH
B2H6 + H2O2/OH
H3O+
CH – O
OH
NaBH4
OH
OH

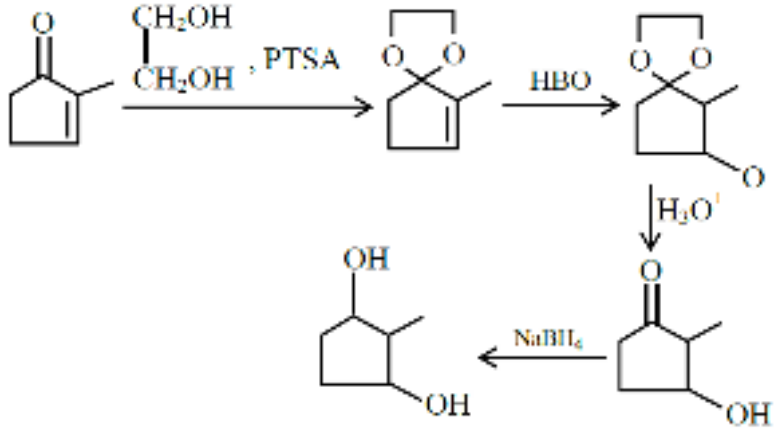
(Q)


O3
Zn
O
CH – O
aq. NaOH
O = C – CH3
CH2OH
CH2OH
, PTSA
B2H6 + H2O2 + OH–
O
O
CH3
OH
H3O+
HO
OH
O
OH
NaBH4

(R)

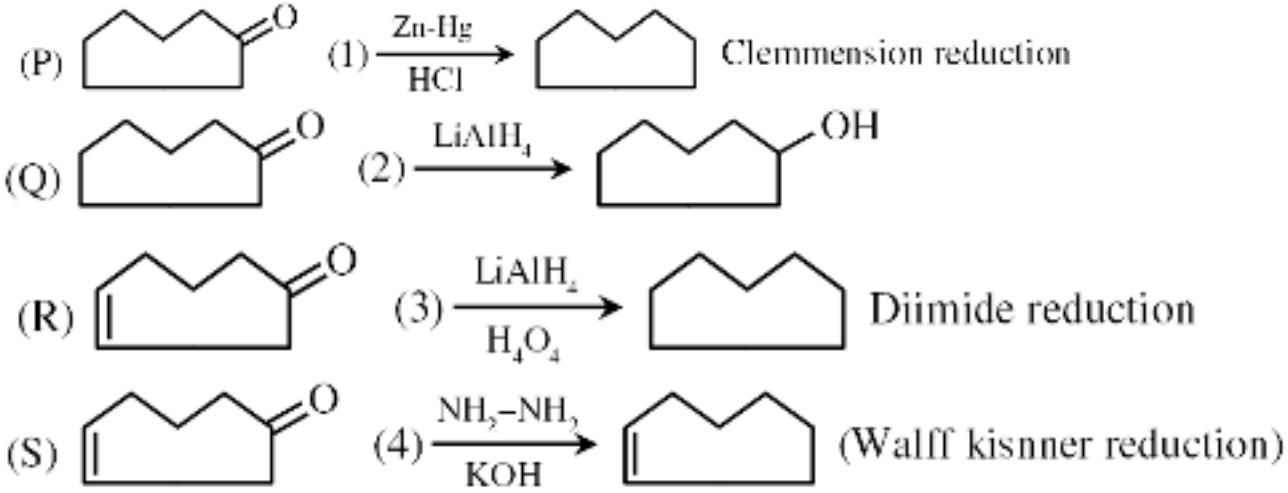

O
CH2OH
CH2OH
, PTSA
O
O
Hg(OAc)2 /H2O
NaBH4
O
O
OH
H3O+
OH
OH
NaBH4
O
OH

(S)



33)

Explaining the Question :-
Question is asking about matching of list I with list II
Concept:-
This question is based on Reduction
Solution :-



Final Answer :- (3)

34) **Aldol reaction** → Must have a H w.r.t carbonyl group off aldehyde & Ketone.
Haloform reaction → Must have $H_3C-\overset{\overset{O}{\|}}{C}-$ group w.r.t. aldehyde and Ketone.
Tollen's Reagent Test → For aldehydes
Cannizaro's Reaction → Aldehydes with no. a H.
Fehling's Test → All aldehydes except PhCH = 0.

PART 3 : MATHEMATICS

35) $\frac{a}{\sqrt{bc}} - 2 = \sqrt{\frac{b}{c}} + \sqrt{\frac{c}{b}}$

$a-2\sqrt{bc}=b+c$

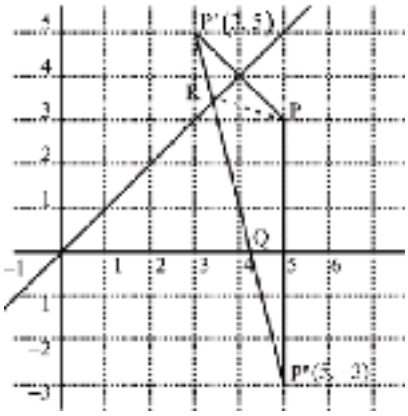
$a=b+c+2\sqrt{bc} \Rightarrow (\sqrt{a})^2=\left(\sqrt{b}+\sqrt{c}\right)^2$

$\Rightarrow \quad \sqrt{a}=\sqrt{b}+\sqrt{c}$ or $-\sqrt{a}=\sqrt{b}+\sqrt{c}$

$\Rightarrow \quad -\sqrt{a}+\sqrt{b}+\sqrt{c}=0$ or $\sqrt{a}+\sqrt{b}+\sqrt{c}=0$

$\therefore$ fixed point is (–1, 1) or (1, 1)

But a, b, c > 0 then (–1, 1) is possible point.

36) 

P' and P" are the images of P w.r.t. y = x and y = 0 respectively. Therefore, P' = (3, 5)

The equation of P'P" is

$$y+3=\frac{5+3}{3-5}(x-5)$$

$$\therefore x=5+3\left(-\frac{1}{4}\right)=\frac{17}{4} \therefore Q\equiv\left(\frac{17}{4},\ 0\right)$$

37) $$x^2\frac{dy}{dx}-2y(x-y)+y^2e^xe^{-x^2/y}=0$$

$$\frac{x^2}{y^2}\frac{dy}{dx}-\frac{2(x-y)}{y}=-e^xe^{-x^2/y}$$

$$e^{x^2/y}\frac{x^2}{y^2}\frac{dy}{dx}-\frac{2x}{y}e^{x^2/y} \quad +2+e^x=0$$

Let $e^{x^2/y}=t$

$$e^{x^2/y}\left(\frac{2xy-x^2\frac{dy}{dx}}{y^2}\right)=\frac{dt}{dx}$$

$$\Rightarrow \quad e^{x^2/y}\frac{2x}{y}-\frac{x^2}{y^2}e^{x^2/y}\frac{dy}{dx}=\frac{dt}{dx}$$

$$\Rightarrow \quad -\frac{dt}{dx}+2t+e^x=0 \Rightarrow \frac{dt}{dx}-2t=e^x$$

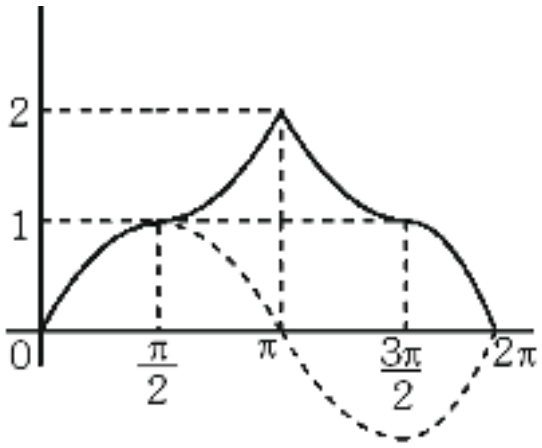
On solving, $t=-e^x+Ce^{2x}$

where C is constant of integration

or $e^{\frac{x^2}{y}-x}=-1+Ce^x$

$$\left(\frac{x^2}{y}-x\right)==\log(Ce^x-1)$$

$x(x-y)=y\log(Ce^x-1)$



38)

f(x) = sin x

f(x) + f(π – x) = 2

$f(x) = 2 - f(\pi - x) = 2 - \sin(\pi - x) = 2 - \sin x \quad x \in \left(\frac{\pi}{2}, \pi\right]$

f(x) = f(2π – x)

f(x + π) = f(π – x)

so curve is symmetric w.r.t line $x = \pi$ for $(\pi, 2\pi]$

$f(x) = f(2\pi - x) = -\sin x$

Area = $2\left(\int_0^{\frac{\pi}{2}} \sin x\,dx + \int_{\frac{\pi}{2}}^{\pi} (2 - \sin x)dx\right)$

$= 2\left(1 + 2 \times \frac{\pi}{2} - 1\right) = 2\pi$

39)

For concurrency $\begin{vmatrix} 1 & \lambda & 6 \\ 2 & 1 & -3 \\ \lambda & 2 & 5 \end{vmatrix} = 0$

$\Rightarrow 3\lambda^2 + 16\lambda - 35 = 0 \Rightarrow (\lambda + 7)(3l - 5) = 0$

$\Rightarrow \lambda = -7, \frac{5}{3} \Rightarrow m = 2$

for not forming triangles either lines are parallel or concurrent

$\left.\begin{array}{l} \frac{1}{2} = \frac{\lambda}{1} \Rightarrow \lambda = 2, \frac{2}{\lambda} = \frac{1}{2} \Rightarrow \lambda = 4 \\ \frac{1}{\lambda} = \frac{\lambda}{2} \Rightarrow \lambda^2 = 2 \Rightarrow \lambda = \pm\sqrt{2} \end{array}\right\} n = 6$

40) $\frac{dy}{dx} + \alpha y = xe^{\beta x}$

$d(e^{\alpha x} y) = xe^{\alpha x} e^{\beta x}$

$d(e^{\alpha x} y) = xe^{(\alpha+\beta)x}$

Case - I : $\alpha + \beta \neq 0$

$d(e^{\alpha x} y) = xe^{(\alpha+\beta)x}$

$e^{\alpha x} y = \frac{xe^{(\alpha+\beta)x}}{(\alpha+\beta)} - \frac{e^{(\alpha+\beta)x}}{(\alpha+\beta)^2} + C$

$$y = \frac{xe^{\beta x}}{(\alpha+\beta)} - \frac{e^{\beta x}}{(\alpha+\beta)^2} + Ce^{-\alpha x}$$

$$\alpha = 1, \beta = 1 \Rightarrow y = \frac{xe^x}{2} - \frac{e^x}{4} + Ce^{-x}$$

as $y(1) = 1 \Rightarrow C = e\left(1 - \frac{e}{4}\right)$

$y(x) = \frac{xe^x}{2} - \frac{e^x}{4} + \left(e - \frac{e^2}{4}\right)e^{-x}$

Case - II : $\alpha + \beta = 0$

$$\Rightarrow \quad \frac{dy}{dx} - \beta y = xe^{\beta x}$$

$d(e^{-\beta x} y) = x$

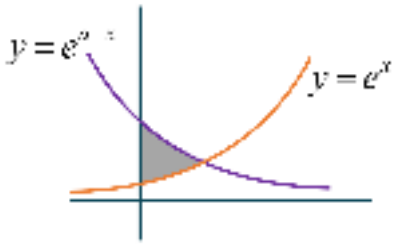
$e^{-\beta x} y = \frac{x^2}{2} + C$

$y = \frac{e^{\beta x} x^2}{2} + Ce^{\beta x}$

$y(1) = 1$

$$\Rightarrow \quad C = \left(1 - \frac{e}{2}\right)\frac{1}{e} \Rightarrow y = e^{\beta x}\frac{x^2}{2} + \left(1 - \frac{e}{2}\right)\frac{1}{e}e^{\beta x}$$

Take $\beta = -1 \Rightarrow y = \frac{x^2}{2}e^{-x} + \left(1 - \frac{e}{2}\right)e^{-x}$.



41)

$$S = \int_0^{a/2} (e^{a-x} - e^x)dx$$

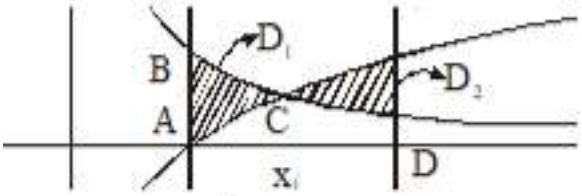
$$= -\left[2e^{\frac{a}{2}} - (e^a + 1)\right] = \left(e^{\frac{a}{2}} - 1\right)^2$$

Now $\lim_{a\to 0} \frac{e^a - 2e^{a/2} + 1}{a^2}$

$$= \lim_{a\to 0}\left(\frac{e^{a/2} - 1}{a/2}\right)^2 \frac{1}{4} = \frac{1}{4}$$

42) Let point of inter section of the curves $y = \frac{1}{x}$ and $y = \ln x$ is x_1

Given $D_1 = D_2$

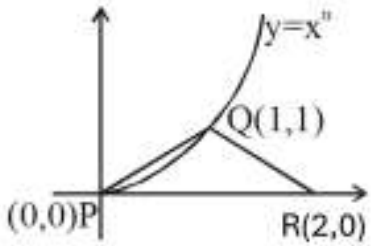


$$\Rightarrow \int_1^{x_1}\left(\frac{1}{x}-\ell nx\right)dx=\int_{x_1}^{a}\left(\ell nx-\frac{1}{x}\right)dx$$

$$\Rightarrow \int_1^{x_1}\left(\frac{1}{x}-\ell nx\right)dx+\int_{x_1}^{a}\left(\frac{1}{x}-\ell nx\right)dx=0$$

$$\Rightarrow \int_1^{a}\left(\frac{1}{x}-\ell nx\right)dx=0 \Rightarrow [\ell n|x|-x\ell nx+x]_1^a=0$$

⇒ (n|a| – a na + a) – (1) = 0
⇒ (1 – a) na = 1 – a ⇒ lna = 1 (if a ≠ 1)
⇒ a = e

43) 

Area of ΔPQR $= \frac{1}{2}\times 2\times 1 = 1$

$$\int_0^1 (x-x^n)\,dx=\frac{3}{10}$$

So,

$$\left(\frac{x^2}{2}-\frac{x^{n+1}}{n+1}\right)_0^1=\frac{3}{10}$$

$$\Rightarrow \frac{1}{2}-\frac{1}{n+1}=\frac{3}{10}$$

$$\Rightarrow \frac{1}{n+1}=\frac{1}{2}-\frac{3}{10}=\frac{1}{5}$$

⇒ n = 4 Ans.

44) $\frac{dy}{dx} = -2x(y-1) \Rightarrow \frac{dy}{dx}+2xy=2x$

Integrating factor = $e^{\int 2x\,dx}=e^{x^2}$

 solution of the d.e. is $ye^{x^2}=\int 2xe^{x^2}dx$

$\Rightarrow ye^{x^2}=e^{x^2}+C$

Given y(0) = 1 ⇒ C = 0 so, $ye^{x^2}=e^{x^2}$

⇒ y = 1

$\therefore \lim_{x\to\infty} y(x) = \lim_{x\to\infty}(1) = 1$

45) $\frac{xdx}{dy} - \frac{x^2}{y} = -y^3$

put $x^2 = t$

$\frac{dt}{dx} - \frac{2}{y}t = -2y^3$

Now solve

46)

$(3x + 7y + 11)\sec\theta + (5x - 3y - 11)\operatorname{cosec}\theta = 0$ passes through intersection of $3x + 7y + 11 = 0$ and $5x - 3y - 11 = 0 \forall$ permissible values of θ given by B(1, –2)

$|PA - PB| \le AB$

$|PA - PB|_{max} = AB = \sqrt{6^2 + 2^2}$

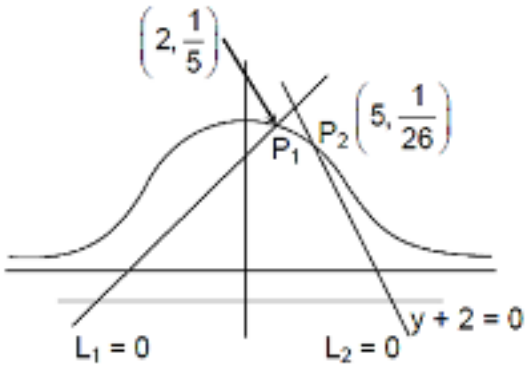
$= 2\sqrt{10} \Rightarrow n = 5$

47) As $P(\alpha, (1 + \alpha^2)^{-1})$ lie on $y = \frac{1}{1 + x^2}$

$\therefore$ On solving $y = \frac{1}{x^2 + 1}$ with L_1, we get $\left(2, \frac{1}{5}\right)$ ...(1)

and with line L_2, we get $P_2\left(5, \frac{1}{26}\right)$ ...(2)

$\therefore$ Equation (1) and (2), we get $2 \le \alpha \le 5$



48) (P) Joint equation is $(x + y - 1)(x - y - 4) = 0$

$x^2 - y^2 = 5x + 3y - 4$

$a + b + c = 5 + 3 - 4 = 4$

(Q) Slope m, m^2

$m + m^2 = 6$

$m^2 + m - 6 = 0$

$(m + 3)(m - 2) = 0$

$m = -3, 2$

$a = m^3$

$a = (-3)^3 = -27$

$a = 2^3 = 8$

(R) We know that joint equation of lines through

(0, 0) $\perp$ to $ax^2 + 2hxy + by^2 = 0$ is $bx^2 - 2hxy + ay^2 = 0$

So for $2x^2 - 5xy + y^2 = 0$

$\perp$ Lines: $x^2 + 5xy + 2y^2 = 0$

$a + 2h + b = 8$

(S) $\tan\theta = \left|\frac{2\sqrt{h^2-ab}}{a+b}\right| = \frac{2\sqrt{\frac{25}{4}-6}}{2+3}$

$\tan\theta = \frac{1}{5} \Rightarrow \theta = \tan^{-1}\frac{1}{5} = \tan^{-1}(m)$

$m = \frac{1}{5} \Rightarrow 5m = 1$

49) (P) $y = e^{4x} + 2e^{-x}$

$\frac{dy}{dx} = 4e^{4x} - 2e^{-x}$

$\frac{d^2y}{dx^2} = 16e^{4x} + 2e^{-x}$

$\frac{d^3y}{dx^3} = 64e^{4x} - 2e^{-x}$

$\therefore \frac{d^3y}{dx^3} - 13\frac{dy}{dx} = \left(64e^{4x} - 2e^{-x}\right) - 13\left(4e^{4x} - 2e^{-x}\right)$

$= 12e^{4x} + 24e^{-x}$

$= 12(e^{4x} + 2e^{-x})$

$= 12y$

$\boxed{K = 12}$ $\boxed{P \to 2}$

(Q) Two lines are possible $\boxed{Q \to 3}$

(R) $2x^4y\frac{dy}{dx} + y^4 = 4x^6$

$y = u^m$

$\frac{dy}{dx} = mu^{m-1}\frac{du}{dx}$

$2x^4u^m mu^{m-1}\frac{du}{dx} + u^{4m} = 4x^6$

$m2x^4u^{2m-1}\left(\frac{du}{dx}\right) + (u)^{4m} = 4\left(x^6\right)$

$\therefore$ 2m – 1 + u = 4m = 6

$\Rightarrow m = \frac{3}{2}$, 2m = 3 $\boxed{R \to 1}$

(S) $y = Ax^m + Bx^{-n}$ is solution of

$x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 12y$ ...(1)

$\frac{dy}{dx} = mAx^{m-1} - nBx^{-n-1}$

$\frac{d^2y}{dx^2} = m(m-1)Ax^{m-2} - n(n+1)Bx^{-n-2}$

$= m(m-1)Ax^{m-2} + n(n+1)Bx^{-n-2}$

$\therefore$ Put in equation (1)

$\Rightarrow x^2(m(m-1)Ax^{m-2} + n(n+1)Bx^{-n-2}) + 2x(mAx^{m-1} - nBx^{-n-1})$

$= 12(Ax^m + Bx^{-m})$

$m(m-1)Ax^m + n(n+1)Bx^{-n} + 2mAx^m - 2nBx^{-n}$

$= 12Ax^m + 12Bx^{-m}$

$\therefore$ m(m – 1)A + 2mA = 12A

$\Rightarrow \boxed{m(m-1)+2m=12} \Rightarrow m=3$

$n(n+1)B - 2nB = 12B$

$\Rightarrow \boxed{n(n+1)-2n=12} \Rightarrow n=-3$

$m+n=0$ $\boxed{S \to 4}$

50) (P) $2x^4ydy + y^2dy = x^2dx - 4x^3y^2dx$

$x^4(2y)dy + y^2(4x^3dx) + y^2dy - x^2dx = 0$

$$d(x^4.y^2) + \frac{1}{3}d(y^3) - \frac{1}{3}d(x^3) = 0$$

$$x^4y^2 + \frac{y^3}{3} - \frac{x^3}{3} = C$$

$$3x^4y^2 + y^3 - x^3 = 3C$$

(Q) Apply componendo & dividendo

$$\frac{e^x}{e^{-x}} = \frac{dx}{dy} \Rightarrow dy = e^{-2x}dx$$

$$y = \frac{-e^{-2x}}{2} + C$$

$2ye^{2x} = -1 + 2Ce^{2x}$

(R) $$x = 1 + xy\frac{dy}{dx} + \frac{\left(xy\frac{dy}{dx}\right)^2}{\underline{|2}} + ...$$

$$x = e^{xy\frac{dy}{dx}}$$

$$\log_e x = xy\frac{dy}{dx}$$

$$\int \frac{\log x}{x}dx = \int ydy$$

$$\frac{\log^2 x}{2} = \frac{y^2}{2} + C$$

$y^2 = \log^2 x + C$

(S) $e^{x^2}ydy + e^{y^2}ydy + e^{x^2}x(y^2-1)dx = 0$

$e^{x^2}2ydy + (y^2-1)e^{x^2}2xdx + e^{y^2}2ydy = 0$

$d\left(e^{x^2}(y^2-1)\right) + d\left(e^{y^2}\right) = 0$

$e^{x^2}(y^2-1) + e^{y^2} = C$

We have,

$$(x-y)f(x+y)-(x+y)f(x-y)=4xy(x^2-y^2)$$

$$=(x^2-y^2)\left[(x+y)^2-(x-y)^2\right]$$

$$=(x-y)(x+y)^3-(x+y)(x-y)^3$$

On comparing, we get

$f(x+y)=(x+y)^3$ and $f(x-y)=(x-y)^3$

Thus, $f(x)=x^3$ *and* $f(y)=y^3$

Now, $\int_0^1 f(x)dx=\int_0^1 x^3dx=\frac{1}{4}\left[x^4\right]_0^1=\frac{1}{4}$

(Q) Cleary, $S_1=y^2-x=0$ .....(i)

And $S_2=x^2+y^2=12$ .....(ii)

On solving Eqs. (i) and (ii), we get $x=3$ *and* $y=\pm\sqrt{3}$

51)

Now, area bounded by the curves S_1 *and* S_2 is $=2\left[\int_0^3\sqrt{x}dx+\int_3^{2\sqrt{3}}\sqrt{12-x^2}dx\right]$

$$=2\left[\frac{2}{3}\left[x^{3/2}\right]_0^3+\frac{1}{2}\left[\left\{\begin{matrix}x\sqrt{\left(2\sqrt{3}\right)^2-x^2}\\+12\sin^{-1}\left(\frac{x}{2\sqrt{3}}\right)\end{matrix}\right\}\right]_3^{2\sqrt{3}}\right]$$

$$=2\left[\frac{2}{3}\left[3\sqrt{3}-0\right]\right]+\frac{1}{2}\left[12\sin^{-1}(1)-\left\{3\sqrt{3}+12\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right\}\right]$$

$$=2\left[2\sqrt{3}+3\pi-\frac{3\sqrt{3}}{2}-6.\frac{\pi}{3}\right]=2\left[\frac{\sqrt{3}}{2}+\pi\right] \qquad =2\pi+\sqrt{3}$$

(R) Required area =Area of circle-Area of squre

$=12\pi-24$

(S) $A=\int_0^1(\sqrt{x}-x)\,dx=\frac{1}{6}$